

Subject Code:
CS505

R22



**C R RAO Advanced Institute of Mathematics,
Statistics and Computer Science (AIMSCS) in
Collaboration with Jawaharlal Nehru Technological
University Hyderabad**



3rd year B.Tech 1st Semester 1st MID Examinations, Aug- 2025

Branch: CSE

Subject: Natural Language Processing

Time: 120 Minutes

DoE: 04-09-2025

Max. Marks: 30 M

PART-A

5X2=10M

Answer all questions

1. Define tokenization. Write about word tokenization.
2. What is morpheme? Write about types of morphemes with example.
3. What is irregularity? Impact of Irregularity on NLP.
4. Explain the Dictionary Lookup model for morphological analysis in NLP. How can a custom dictionary be created? Illustrate with an example.
5. Write about NLTK.

PART – B

4X5 = 20 M

Answer any FOUR questions

1. Why is document structure important in NLP? Discuss the different methods used to identify the structure of a document.
2. Explain the difference between stemming and lemmatization with examples.
3. Compare constituency parsing with dependency parsing methods with examples.
4. Describe shift-reduce parsing techniques. Solve Consider the Grammar:

$E \rightarrow E+T \mid T$

$T \rightarrow T * F \mid F$

$F \rightarrow id$

Perform Shift Reduce parsing for the input string "id + id + id".

5. Explain Top-Down and Bottom-Up parsing techniques with examples.
6. Explain about various phases of NLP.

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3rd year B.Tech 1st Semester 2nd MID Examinations, Nov- 2025

Branch: CSE

Subject: Natural Language Processing
Time: 120 Minutes

DoE: 07-11-2025
Max. Marks: 30 M

PART-A

Answer all questions

5X2=10M

1. Write about types of semantic parsing techniques with advantage and disadvantages
2. Write about n-gram language models.
3. Write about word sense
4. What is predicate argument structure and its goal. Why it matters in NLP?
5. Write challenges in Language Modeling.

PART – B

Answer any **FOUR** questions

4X5 = 20 M

1. Explain about Word Sense Disambiguation(WSD) and various approaches to WSD.
2. Explain Extended Lesk Algorithm with example evaluation.
3. Explain difference between FrameNet and PropBank
4. Write about Meaning Representation Systems and its types in detail.
5. Write about system paradigms
6. Write about Probabilistic Context Free Grammar (PCFG) model for Resolutions of ambiguity parsing, and solve below using grammar

$S \rightarrow NP, VP$ [1.0], $VP \rightarrow V, NP$ [0.7],

$VP \rightarrow VP, PP$ [0.3], $NP \rightarrow Det, N$ [0.6],

$NP \rightarrow N, PP$ [0.4], $PP \rightarrow P, N$ [1.0],

$Det \rightarrow the$ [1.0], $N \rightarrow man$ [0.5],

$N \rightarrow telescope$ [0.5], $NP \rightarrow I$ [1.0],

$V \rightarrow saw$ [1.0], $P \rightarrow with$ [1.0]

Target sentence: "I saw the man with the telescope"